

1) Simple Calculator

Description: Create a C++ program that takes two floating-point numbers and an operator character (+, -, *, /) as input. Perform the corresponding arithmetic operation and display the result. Handle division by zero.

Example Input:

10.5
5.0
+

Example Output: Result: 15.5

Example Input:

20.0
0.0
/

Example Output: Error: Division by zero is not allowed.

2) Leap Year Checker

Description: Write a C++ program that takes a year (integer) as input and determines if it's a leap year.

A year is a leap year if it is divisible by 4, unless it is divisible by 100 but not by 400.

Example Input: 2000

Example Output: 2000 is a leap year.

Example Input: 1900

Example Output: 1900 is not a leap year.

Example Input: 2024

Example Output: 2024 is a leap year.

3) Discount Calculator

Description: A store offers discounts based on the total purchase amount and whether the customer is a premium member.

- If total purchase amount is less than \$100: No discount.
- If total purchase amount is \$100 or more:
 - If a premium member: 15% discount.
 - If not a premium member: 10% discount.

Example Input:

75.0

Example Output: Final amount: \$75.0

Example Input:

120.0

Example Output: Final amount: \$102.0

4) Electricity Bill Calculation

Description: Write a C++ program to calculate a consumer's electricity bill based on their units consumed. The tariff rates are tiered:

- **Up to 50 units:** \$0.50 per unit
- **Next 100 units (51 to 150 units):** \$0.75 per unit
- **Next 100 units (151 to 250 units):** \$1.20 per unit
- **Above 250 units:** \$1.50 per unit

A **fixed meter charge of \$50** is added to all bills. Additionally, a **surcharge of 20%** is applied to the **total bill amount (including meter charge, but before surcharge)** if this amount exceeds \$400.

Your program should:

1. Prompt the user to enter the total units consumed.
2. Calculate the electricity charge based on the slab rates.
3. Add the fixed meter charge.
4. Apply the 20% surcharge if the conditions are met.
5. Display the total units consumed, the electricity charge, the meter charge, the surcharge amount (if any), and the final bill amount. Format all monetary values to two decimal places.

Example Input: 120

Example Output:

Total Units Consumed: 120

Electricity Charge: \$ 77.50

Meter Charge: \$ 50.00

Surcharge (20%): \$ 0.00

Final Bill Amount: \$ 127.50

Example Input: 400

Example Output:

Total Units Consumed: 400

Electricity Charge: \$ 445.00

Meter Charge: \$ 50.00

Surcharge (20%): \$ 99.00

Final Bill Amount: \$ 594.00